HW #3

ALGEBRAIC NUMBER THEORY, GU4043; INSTRUCTOR: GYUJIN OH

Question 1.

(1) Let K be a field. Show that $K[X]$ is a Dedekind domain.

Hint. You can use that $K[X]$ is a UFD.

(2) Let K be a field. Show that $K[X, Y]$ is **not** a Dedekind domain.

(3) Let R be a Dedekind domain, and let $r \in R$ be a non-zero element. Show that the ring

$$R\left[\frac{1}{r}\right] := \left\{\frac{a}{r^n} \in \mathrm{Frac}(R) \ : \ a \in R, n \geq 0\right\}$$

is also a Dedekind domain.

Proof.

(1) As $K[X]$ is a UFD, it is normal. It is standard that $K[X]$ is Noetherian. Thus, we only need to show that $\dim K[X] = 1$, i.e., any nonzero prime ideal is maximal. Let $\mathfrak{p} \subset K[X]$ be a nonzero prime ideal. As $\mathfrak{p}$ is nonzero, there is a nonzero element $f(X) \in \mathfrak{p}$, which implies that $\mathfrak{p} \supset (f(X))$. Therefore, $K[X]/\mathfrak{p}$ is a quotient of $K[X]/(f(X))$, which is finite-dimensional as a vector space over K. Therefore, $K[X]/\mathfrak{p}$ is also finite-dimensional as a K-vector space. As $K[X]/\mathfrak{p}$ is an integral domain, by Exercise A.3, $K[X]/\mathfrak{p}$ is a field, which implies that $\mathfrak{p}$ is a maximal ideal.

(2)

$$(X, Y) \supsetneq (X) \supsetneq (0)$$

It is easy that (X, Y) and (X) are prime ideals: $K[X, Y]/(X, Y) \cong K$ and $K[X, Y]/(X) \cong K[Y]$ are integral domains.

(3) Let us first try to figure out the ideals of $R[1/r]$. If $I \subset R[1/r]$ is an ideal, then $J := I \cap R$ is an ideal of R. Let I' be the ideal of $R[1/r]$ generated by J. By definition, $I \supset I'$. We claim that $I = I'$. If there is $x \in I$, then $x = \frac{a}{r^n}$ for some $a \in R$ and $n \geq 0$. Thus, $a = r^n x \in I$, which is also an element of R, so $a \in J$. Thus, I' contains $x = \frac{a}{r^n}$. This implies that any ideal of $R[1/r]$ is generated by an ideal of R. This easily implies that $R[1/r]$ is Noetherian (i.e., every ideal is finitely generated).

We now show that $R[1/r]$ is normal. Suppose that $\alpha \in \mathrm{Frac}(R)$ is integral over $R[1/r]$. Then, there exists a monic polynomial $f(X) \in R[1/r][X]$ where $f(\alpha) = 0$, i.e., there is a relation

$$\alpha^m + \frac{a_{m-1}}{r^{n_{m-1}}} a^{m-1} + \cdots + \frac{a_1}{r^{n_1}} a + \frac{a_0}{r^{n_0}} = 0, \qquad a_{m-1}, \cdots, a_0 \in R, \quad n_{m-1}, \cdots, n_0 \geq 0.$$

At the expense of increasing n_i's and multiplying a_i's with powers of r's, we can assume that $n_{m-1} = \cdots = n_0 = N$. Thus we have

$$r^N \alpha^m + a_{m-1}\alpha^{m-1} + \cdots + a_1 \alpha + a_0 = 0.$$

After multiplying everything with $r^{N(m-1)}$, we get

$$r^{Nm}\alpha^m + a_{m-1}r^{N(m-1)}\alpha^{m-1} + \cdots + a_1 r^{N(m-1)}\alpha + a_0 r^{N(m-1)} = 0,$$

or

$$(r^N\alpha)^m + a_{m-1}(r^N\alpha)^{m-1} + a_{m-2}r^N(r^N\alpha)^{m-2} + \cdots + a_1 r^{N(m-2)}(r^N\alpha) + a_0 r^{N(m-1)} = 0.$$

Thus, $r^N\alpha$ is integral over R. As R is normal, $r^N\alpha \in R$. Thus, $\alpha \in R[1/r]$, which implies that $R[1/r]$ is normal.

Now we show that $\dim R[1/r] = 1$. Note that, by the definition of prime ideals, if $\mathfrak{p}$ is a prime ideal of $R[1/r]$, $\mathfrak{p} \cap R$ is either equal to R or a prime ideal of R. Suppose on the contrary that there is a nonzero prime ideal $\mathfrak{p}$ of $R[1/r]$ that is not maximal. Then, there is a prime ideal $(0) \subsetneq \mathfrak{p} \subsetneq \mathfrak{q} \subsetneq R[1/r]$. Note that each of $\mathfrak{p} \cap R$ and $\mathfrak{q} \cap R$ is either equal to R or a prime ideal of R, and $\mathfrak{p} \cap R \subset \mathfrak{q} \cap R$. As $\dim R = 1$, there are three possibilities:

- $\mathfrak{p} \cap R = \mathfrak{q} \cap R$. However, this is impossible, as we already observed that any ideal I of $R[1/r]$ is generated by $I \cap R$, so $\mathfrak{p} \cap R = \mathfrak{q} \cap R$ would imply that $\mathfrak{p}$ and $\mathfrak{q}$ are the same ideals.
- $\mathfrak{p} \cap R = (0)$. Again, this is impossible, as by the same reason as above, $\mathfrak{p}$ would be generated by 0, which would mean that $\mathfrak{p}$ is the zero ideal.
- $\mathfrak{q} \cap R = R$. Again, this is impossible, as by the same reason as above, $\mathfrak{q}$ would be generated by R, which would mean that $\mathfrak{q} = R[1/r]$, which is not a prime ideal.

Therefore, the assumption is wrong, and we have deduced that $\dim R[1/r] = 1$.

□

Question 2. Let $K = \overline{K}$ be an algebraically closed field, and let $f(X) \in K[X]$. We are interested in when the ring[1]

$$R := K[X,Y]/(Y^2 - f(X))$$

is a Dedekind domain. It is clear that $f(X)$ needs to be a non-square for R to be an integral domain, which we assume.

(1) If R is a Dedekind domain, show that the following condition is satisfied:

$$(\star) \qquad \text{There is no element } \alpha \in K \text{ such that } 2f(\alpha) = \frac{df}{dX}(\alpha) = 0.$$

Hint. Suppose that there is such an α. Let $\beta \in K$ be such that $\beta^2 = f(\alpha)$. Show that $\frac{Y-\beta}{X-\alpha} \in \mathrm{Frac}(R)$ is integral over R, and that this element is not an element of R. **Treat the cases** $\mathrm{char}\, K \neq 2$ **and** $\mathrm{char}\, K = 2$ **separately.**

(2) Show that, when $\mathrm{char}\, K \neq 2$, $(\star)$ is just merely requiring that $f(X)$ is a square-free polynomial. Why is $(\star)$ a much more stringent condition when $\mathrm{char}\, K = 2$?

(3) If R satisfies $(\star)$, show that R is a Dedekind domain.

Hint. If $\mathrm{char}\, K = 2$, show that R is, as an abstract ring, isomorphic to $K[X]$ (!). If $\mathrm{char}\, K \neq 2$, mimic the computation of ring of integers for quadratic fields.

Proof.

(1) We treat the cases $\mathrm{char}\, K \neq 2$ and $\mathrm{char}\, K = 2$ separately.

[1] This is the analogue of the "quadratic field" case for $K[X]$ instead of $\mathbb{Q}$.

- $\operatorname{char} K \neq 2$. In this case, the condition ($\star$) becomes that there is no $\alpha \in K$ such that $f(\alpha) = \frac{df}{dX}(\alpha) = 0$. Suppose the contrary that there is such an α. Firstly, $f(\alpha) = 0$ means that $(X-\alpha)|f(X)$. Let $f(X) = g(X)(X-\alpha)$. Then, $\frac{df}{dX} = \frac{dg}{dX}(X-\alpha)+g(X)$, so $\frac{df}{dX}(\alpha) = g(\alpha)$. Therefore, $\frac{df}{dX}(\alpha) = 0$ implies that $(X-\alpha)|g(X)$, or in turn, $(X-\alpha)^2|f(X)$. Let $f(X) = (X-\alpha)^2 h(X)$ for $h(X) \in K[X]$. Consider the element
$$\gamma := \frac{Y}{X-\alpha} \in \operatorname{Frac}(R).$$
Note that
$$\gamma^2 = \frac{Y^2}{(X-\alpha)^2} = \frac{f(X)}{(X-\alpha)^2} = h(X),$$
so γ is a root of the monic polynomial $i(T) \in R[T]$ given by $i(T) := T^2 - h(X)$. This implies that γ is integral over R. As R is a Dedekind domain, $\gamma \in R$. However, it is clear that the elements of R are of the form $a(X) + b(X)Y$ for $a(X), b(X) \in K[X]$, whereas the elements of $\operatorname{Frac}(R)$ are of the form $a(X) + b(X)Y$ for $a(X), b(X) \in K(X) := \operatorname{Frac}(K[X])$. Therefore, $\gamma \notin R$, a contradiction.
- $\operatorname{char} K = 2$. In this case, the condition ($\star$) becomes that there is no $\alpha \in K$ such that $\frac{df}{dX}(\alpha) = 0$. Suppose the contrary that there is such an α. Now $f(\alpha)$ may not be zero. As K is algebraically closed, one can still find $\beta \in K$ such that $\beta^2 = f(\alpha)$. Consider the element
$$\gamma := \frac{Y-\beta}{X-\alpha} \in \operatorname{Frac}(R).$$
Note that
$$\gamma^2 = \frac{(Y-\beta)^2}{(X-\alpha)^2} = \frac{Y^2-\beta^2}{(X-\alpha)^2} = \frac{f(X)-f(\alpha)}{(X-\alpha)^2},$$
where the identity holds as $\operatorname{char} K = 2$. Let $g(X) := f(X) - f(\alpha)$. As $g(\alpha) = 0$, $(X-\alpha)|g(X)$. Let $g(X) = (X-\alpha)h(X)$ for some $h(X) \in K[X]$. Then, as $\frac{dg}{dX} = \frac{df}{dX}$, we have $\frac{dg}{dX}(\alpha) = 0$, which implies similarly that $(X-\alpha)|h(X)$, or in turn $(X-\alpha)^2|g(X)$. Thus $g(X) = (X-\alpha)^2 i(X)$ for some $i(X) \in K[X]$. Thus
$$\gamma^2 = i(X).$$
Therefore, γ is a root of the monic polynomial $j(T) \in R[T]$ given by $j(T) := T^2 - i(X)$. This implies that γ is integral over R, and as R is a Dedekind domain, $\gamma \in R$. However, by the same reason as above, $\gamma \notin R$, so a contradiction.

(2) For $\operatorname{char} K \neq 2$, this is what we observed already. For $\operatorname{char} K = 2$, this is saying that $\frac{df}{dX}$ is a polynomial that has no root in K. As K is algebraically closed, this means that $\frac{df}{dX}$ is a nonzero constant polynomial, so the form of $f(X)$ is extremely constrained; namely, ($\star$) holds if and only if $f(X) = aX + \sum_{n=0}^{M} b_n X^{2n}$ for some $a \neq 0$ and $b_0, \cdots, b_M \in K$.

(3) If $\operatorname{char} K = 2$, we know that
$$f(X) = aX + \sum_{n=0}^{M} b_n X^{2n}.$$

For $n = 0, \cdots, M$, let $c_n \in K$ be such that $c_n^2 = b_n$ (this is possible as K is algebraically closed). Then, as $\operatorname{char} K = 2$,

$$\left(\sum_{n=0}^{M} c_n X^n\right)^2 = \sum_{n=0}^{M} c_n^2 X^{2n} = \sum_{n=0}^{M} b_n X^{2n}.$$

Therefore, $f(X) = aX + g(X)^2$ for some $g(X) \in K(X)$. Therefore,

$$\begin{aligned} R &= K[X,Y]/(Y^2 - f(X)) = K[X,Y]/(Y^2 - aX - g(X)^2) \\ &= K[X,Y]/((Y - g(X))^2 - aX) = K[X,Z]/(Z^2 - aX), \end{aligned}$$

where $Z := Y - g(X)$. Taking $b \in K$ such that $b^2 = a$ and using $W := Z/b$, we get

$$R = K[X,Z]/(Z^2 - aX) = K[X,W]/(b^2W^2 - aX) = K[X,W]/(aW^2 - aX) = K[X,W]/(W^2 - X).$$

It is clear that $K[X,W]/(W^2 - X) \cong K[W]$ (you just name W^2 as X, and having them as an additional generator does not add anything new to $K[W]$). Therefore, R is abstractly isomorphic to $K[X]$, which is a Dedekind domain.

If $\operatorname{char} K \neq 2$, it is still immediate that R is Noetherian, as $K[X]$ is Noetherian and R is finitely generated over $K[X]$. We then show that R is normal. By Lemma 2.2 (or its proof), this is the same as showing that R is the integral closure of $K[X]$ in $\operatorname{Frac}(R)$. Note that a general element of $\operatorname{Frac}(R)$ is of the form $a(X) + b(X)Y$ for $a(X), b(X) \in K(X)$. Suppose that this is integral over $K[X]$. If $b(X) = 0$, then this means that $a(X) \in K(X)$ is integral over $K[X]$, which, by the normality of $K[X]$, means that $a(X) \in K[X]$, so that $a(X) + b(X)Y = a(X) \in R$. Therefore, we can assume that $b(X) \neq 0$. Then, the minimal polynomial of the element over $K(X)$ is

$$T^2 - \operatorname{Tr}_{\operatorname{Frac}(R)/K(X)}(a(X) + b(X)Y)T + N_{\operatorname{Frac}(R)/K(X)}(a(X) + b(X)Y) \in K(X)[T].$$

By the Gauss's lemma, this must be in $K[X][T]$ (and this is if-and-only-if). Note that

$$\begin{aligned} \operatorname{Tr}_{\operatorname{Frac}(R)/K(X)}(a(X) + b(X)Y) &= 2a(X) \\ N_{\operatorname{Frac}(R)/K(X)}(a(X) + b(X)Y) &= a(X)^2 - f(X)b(X)^2. \end{aligned}$$

As $\operatorname{char} K \neq 2$, $\operatorname{Tr}_{\operatorname{Frac}(R)/K(X)}(a(X) + b(X)Y) = 2a(X) \in K[X]$ means $a(X) \in K[X]$. Thus, $N_{\operatorname{Frac}(R)/K(X)}(a(X) + b(X)Y) = a(X)^2 - f(X)b(X)^2 \in K[X]$ implies that $f(X)b(X)^2 \in K[X]$. As $(\star)$ holds, $f(X)$ is square-free. Therefore, $b(X) \in K[X]$. This implies that $a(X) + b(X)Y \in R$. This shows that R is the integral closure of $K[X]$ in $\operatorname{Frac}(R)$.

It remains to show that $\dim R = 1$. Let $\mathfrak{p} \subset R$ be a nonzero prime ideal. As we have seen in Question 1, $\mathfrak{p} \cap K[X]$ is either equal to $K[X]$ or a prime ideal of $K[X]$. We use that $\dim K[X] = 1$ to reduce to the following three possibilities.

- If $\mathfrak{p} \cap K[X] = K[X]$, then $\mathfrak{p}$ contains the ideal of R generated by $K[X]$. However, as $1 \in K[X]$, this implies $\mathfrak{p} = R$, which is a contraidction.
- If $\mathfrak{p} \cap K[X] = (0)$, then we claim that $\mathfrak{p} = (0)$. Suppose not. Then, there exists a nonzero element $a(X) + b(X)Y \in \mathfrak{p}$, $a(X), b(X) \in K[X]$. Thus $(a(X) + b(X)Y)(a(X) - b(X)Y) \in \mathfrak{p}$, so $a(X)^2 - b(X)^2 f(X) \in \mathfrak{p}$, which is also an element of $K[X]$. This implies that $a(X)^2 - b(X)^2 f(X) = 0$, or $a(X)^2 = b(X)^2 f(X)$. On the other hand, as $f(X)$ is square-free, this implies that $f(X)$ must have no nonconstant factor (otherwise it would contradict the UFD property of $K[X]$). This contradicts the assumption that $f(X)$ is not a square. Therefore, $\mathfrak{p} \cap K[X] \neq (0)$.

- If $\mathfrak{p} \cap K[X]$ is a maximal ideal, then K being algebraically closed implies that $\mathfrak{p} \cap K[X] = (X - \alpha)$ for some $\alpha \in K$. Note that the ring homomorphism

$$K \cong K[X]/(X - \alpha) = K[X]/(\mathfrak{p} \cap K[X]) \to R/\mathfrak{p}$$

is well-defined and injective. Therefore, $R/\mathfrak{p}$ is naturally a K-algebra. As $\{1, Y\}$ is a $K[X]$-basis of R, it follows that $\{1, Y\}$ will generate $R/\mathfrak{p}$ as a K-algebra as well. This implies that $R/\mathfrak{p}$ is, as a K-vector space, of dimension ≤ 2. As $R/\mathfrak{p}$ is an integral domain, by Exercise A.3, $R/\mathfrak{p}$ is a field, which implies that $\mathfrak{p}$ is a maximal ideal.

Therefore, any nonzero prime ideal of R is maximal, so $\dim R = 1$.

□

Question 3. Recall that, in class, we showed that the unique prime factorization of numbers does not hold for $\mathbb{Z}[\sqrt{-5}]$, because of the identity

$$6 = 2 \cdot 3 = (1 + \sqrt{-5}) \cdot (1 - \sqrt{-5}).$$

We also saw how this is salvaged by considering ideals. The key was that there are prime ideals of the Dedekind domain $\mathbb{Z}[\sqrt{-5}]$ that are not principal. See Example 2.10 for a detailed explanation.

We will call the identity of the above form a **counterexample to unique prime factorization of numbers** for $R = \mathbb{Z}[\sqrt{-5}]$. Namely, it is an identity of the form $ab = cd$ where a, b, c, d are irreducible elements of R such that a or b is not equal to c or d even up to multiplication by a unit.

Here comes the task: go to LMFDB (`https://lmfdb.org`), and in the "Number fields" section, choose your favorite number field K which is

- not $\mathbb{Q}(\sqrt{-5})$,
- and has a non-trivial class group (i.e., the "Class group" column is not "trivial").

For such a K, find a counterexample to unique prime factorization of numbers for $\mathcal{O}_K$, as defined above. Explain also why this is salvaged by the unique factorization of prime ideals, just as we did in class.

Proof. For example, $K = \mathbb{Q}(\sqrt{-14})$ has a non-trivial class group, and has

$$18 = 2 \cdot 3 \cdot 3 = (2 + \sqrt{-14})(2 - \sqrt{-14}).$$

We show that $2, 3, 2 \pm \sqrt{-14}$ are irreducible elements in $\mathcal{O}_K = \mathbb{Z}[\sqrt{-14}]$.

- 2: if $x + y\sqrt{-14} \in \mathbb{Z}[\sqrt{-14}]$ divides 2, then $N(x + y\sqrt{-14}) = x^2 + 14y^2$ divides $N(2) = 4$, so $x^2 + 14y^2$ is either 1, 2, or 4. This implies that $y = 0$ (otherwise $x^2 + 14y^2 \geq 14$), and $x = \pm 1$ or ± 2. Thus 2 is irreducible.
- 3: if $x + y\sqrt{-14} \in \mathbb{Z}[\sqrt{-14}]$ divides 3, then $N(x + y\sqrt{-14}) = x^2 + 14y^2$ divides $N(3) = 9$, so $x^2 + 14y^2$ is either 1, 3, or 9. This implies that $y = 0$ (otherwise $x^2 + 14y^2 \geq 14$), and $x = \pm 1$ or ± 3. Thus 3 is irreducible.
- $2 \pm \sqrt{-14}$: if $x + y\sqrt{-14} \in \mathbb{Z}[\sqrt{-14}]$ divides $2 \pm \sqrt{-14}$, then $N(x + y\sqrt{-14}) = x^2 + 14y^2$ divides $N(2 \pm \sqrt{-14}) = 18$, so $x^2 + 14y^2$ is one of the following: $1, 2, 3, 6, 9, 18$. This implies that y is either 0 or ± 1 (otherwise $x^2 + 14y^2 \geq 14 \cdot 4 = 56$). If $y = 0$, then x must be either ± 1 or ± 3. However, ± 3 does not divide $2 \pm \sqrt{-14}$, as $\frac{2}{3} \pm \frac{1}{3}\sqrt{-14} \notin \mathbb{Z}[\sqrt{-14}]$. If $y = \pm 1$, then $x^2 + 14y^2$ must be 18. Therefore, $\frac{2 \pm \sqrt{-14}}{x + y\sqrt{-14}}$ must be of norm 1 and is an element of $\mathcal{O}_K$, which means that it is a unit. Therefore, $2 \pm \sqrt{-14}$ is irreducible.

This is salvaged by the following identities

$$(2) = (2, 2+\sqrt{-14})(2, 2-\sqrt{-14})$$

- Check:

$$\begin{aligned}
(2, 2+\sqrt{-14})(2, 2-\sqrt{-14}) &= (4, 4+2\sqrt{-14}, 4-2\sqrt{-14}, (2+\sqrt{-14})(2-\sqrt{-14})) \\
&= (4, 2\sqrt{-14}, 18) \\
&= (2, 2\sqrt{-14}) \\
&= (2)
\end{aligned}$$

$$(3) = (3, 2+\sqrt{-14})(3, 2-\sqrt{-14})$$

- Check:

$$\begin{aligned}
(3, 2+\sqrt{-14})(3, 2-\sqrt{-14}) &= (9, 6+3\sqrt{-14}, 6-3\sqrt{-14}, (2+\sqrt{-14})(2-\sqrt{-14})) \\
&= (9, 6+3\sqrt{-14}, 6-3\sqrt{-14}, 18) \\
&= (9, 6+3\sqrt{-14}, 6-3\sqrt{-14}) \\
&= (9, 12, 6-3\sqrt{-14}) \\
&= (3, 6-3\sqrt{-14}) \\
&= (3)
\end{aligned}$$

$$(2+\sqrt{-14}) = (2, 2+\sqrt{-14})(3, 2+\sqrt{-14})^2$$

- Check:

$$\begin{aligned}
(3, 2+\sqrt{-14})^2 &= (9, 6+3\sqrt{-14}, (2+\sqrt{-14})^2) \\
&= (9, 6+3\sqrt{-14}, -10+4\sqrt{-14}) \\
&= (9, 6+3\sqrt{-14}, -16+\sqrt{-14}) \\
&= (9, 54, -16+\sqrt{-14}) \\
&= (9, -16+\sqrt{-14}) \\
&= (9, 2+\sqrt{-14})
\end{aligned}$$

and

$$
\begin{aligned}
(2, 2+\sqrt{-14})(3, 2+\sqrt{-14})^2 &= (2, 2+\sqrt{-14})(9, 2+\sqrt{-14}) \\
&= (18, 4+2\sqrt{-14}, 18+9\sqrt{-14}, (2+\sqrt{-14})^2) \\
&= (18, 4+2\sqrt{-14}, 9\sqrt{-14}, -10+4\sqrt{-14}) \\
&= (18, 4+2\sqrt{-14}, 9\sqrt{-14}, -18) \\
&= (18, 4+2\sqrt{-14}, 9\sqrt{-14}) \\
&= (18, 4+2\sqrt{-14}, -16+\sqrt{-14}) \\
&= (18, 36, -16+\sqrt{-14}) \\
&= (18, -16+\sqrt{-14}) \\
&= (18, 2+\sqrt{-14}) \\
&= ((2+\sqrt{-14})(2-\sqrt{-14}), 2+\sqrt{-14}) \\
&= (2+\sqrt{-14})
\end{aligned}
$$

$$(2-\sqrt{-14}) = (2, 2-\sqrt{-14})(3, 2-\sqrt{-14})^2$$

The last identity is proved exactly in the same way as the the third identity with only the signs changed. $\square$